

Deliverable 2: Data collection method for DASC4300 Project
Project Title: The Impact of Sleep Cycles on Productivity and Well-Being

Group Information:

- **Group Name:** Better Sleep, Energized Life
- **Group Member:** xxx

In this deliverable, we are going to create a program that is able to recreate the dataset.

Data Collection Method

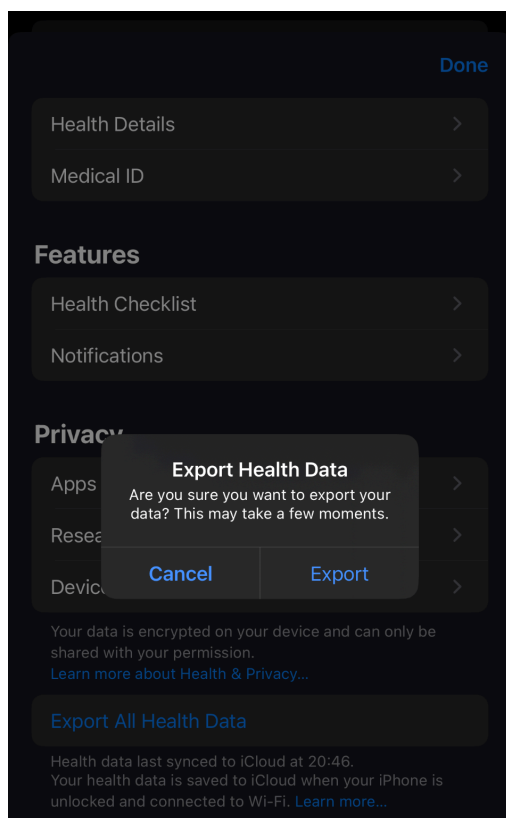
Looking at our dataset, the last few columns consist of data based on personal experience and subjective feelings (productivity score, stress score and mood score), so making a GUI program for respondents to input is a must. However, a GUI program is not sufficient for our dataset, since there is one column that is impossible for the users to provide themselves, which is 'Start Time' (of a sleep). Therefore we also need to include a sleep-tracking device in our data collection.

Technology Explanation

The Apple Watch tracks our sleep process and stages primarily by using its built-in accelerometer to monitor motion. During sleep, our body naturally becomes less active, and the watch detects patterns of movement, such as tossing and turning or periods of stillness, to estimate when you're asleep and what stage of sleep you might be in. Light sleep typically involves more movement, while deep sleep is characterized by little to no motion. The watch combines this motion data with other metrics, like heart rate and breathing rate, using advanced algorithms and machine learning to provide a more accurate picture of your sleep. Although it doesn't measure brain activity like medical-grade devices, it offers a convenient way to estimate sleep stages, total sleep time, and trends over time, helping us to track the sleeping time.

Guidance on Data Collection Process

An apple watch and iphone will be lend to the participants for this data collection process. The user would need to wear the apple watch while sleeping. At the same time, participants also need to make a record of their sleeping quality, exercise time, working hours, productivity score, mood score and stress level each day of the assessment period. After that, they will need to generate a health report in the 'health' app in their iphones, and open the zip file in the computers. It will include an 'export.xml' file, which they will need to upload when they are filling in the questionnaire from the program.



For the data inputs, I have set the data type of each input matching the datatype of that in my dataset so that if the user has typed used the wrong datatype or wrong value, an error message will pop up.

Basic Information

Age:

Gender:

Person ID:

Number of Days Participating:

Next



Please enter the four-digit ID we have provided.

OK

Day 1 of 3

Date: - -

Caffeine Intake:

Sleep Quality (1-10):



Please enter a valid date.

OK

For caffeine intake, there is no other way to keep track except asking them what have they drank that day. In the questionnaire, the users would need to click ‘add drink’ if they have drank any caffeinated drinks that day. For coffee, they would need to choose the coffee type and then input the volume of the coffee consumed. For energy drink, they would need to choose the energy drink type and then input the amount of servings consumed.

Caffeine Intake:

Drink Type:

Drink Type:

After they have uploaded the xml file in the last page of the form and clicked ‘submit’, the sleep data will be extracted from the program with the xml file, and the sleep data will be matched to the data input from the form by the same date (of end sleep, because we want to understand what will a sleep affect the productivity of a person the NEXT day). The data input will be converted to our csv file.

GUI_for Updating_Deliverable2.py | Deliverable2_20853269.py | Deliverable2_update_test.py

output_sleep_cycle_productivity.csv

1	Date,Person_ID,Age,Gender,Sleep Start Time,Sleep End Time,Total Sleep Hours,Sleep Quality
2	2025-03-20,4300,22,Male,4.24,9.83,5.58,1,1,109.0,1,1,1,1,1
3	2025-03-20,4300,22,Male,4.24,9.83,5.58,1,1,395.0,1,1,1,1,1
4	2025-03-21,4300,22,Male,4.47,10.52,6.05,2,2,0,2,2,2,2,2
5	2025-03-22,4300,22,Male,3.64,12.03,8.38,3,3,0,3,3,3,3,3
6	